

Title of the Project: MedREX: Revolutionizing Medical Equipment Procurement

Group Name: MedRex

Group Members:

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Introduction

MedRex is an innovative web based centralized marketplace designed to facilitate the procurement and distribution of medical equipment in a structured and efficient digital environment. The platform aims to connect healthcare institutions such as clinics, hospitals, and diagnostic laboratories with verified medical equipment suppliers, ensuring transparency, reliability, and accessibility in the purchasing process. By maintaining a comprehensive database of medical equipment sourced from different manufacturers worldwide, MedRex provides users with a well-organized and detailed view of product specifications, pricing, and availability. The system allows users to create secure accounts through which they can manage their activity, including viewing previous search history and marking specific products as favorites for future reference. In addition, MedRex incorporates an advanced comparison feature that enables users to select multiple products up to four at a time and evaluate them side by side based on key attributes such as specifications, functionality, and price. This feature supports informed decision-making in critical procurement scenarios. Furthermore, verified medical equipment retailers are able to register as “Sellers” on the platform. These sellers can list products as available for sale within their respective inventories, along with associated pricing and availability information. Seller details, including store name and address, are displayed alongside product listings to ensure transparency and trustworthiness. Overall, MedRex serves as a unified digital ecosystem that enhances efficiency, accuracy, and trust in the medical equipment procurement process.

Motivation

Medical equipment plays a critical role in modern healthcare systems, directly influencing the quality of diagnosis, treatment, and patient care. In many developing regions, including Bangladesh, healthcare institutions often face significant challenges in sourcing reliable medical equipment due to fragmented supply channels, inconsistent pricing, and limited transparency in product availability. As a result, clinics, hospitals, and diagnostic laboratories frequently struggle to identify the most suitable equipment that meets both their technical requirements and budget constraints. Although numerous suppliers and distributors exist in the market, the absence of a unified platform makes it difficult for healthcare providers to compare products efficiently or verify seller credibility. This fragmentation leads to inefficiencies, delays in procurement, and increased operational costs.

The motivation behind MedRex is to address these challenges by creating a centralized digital marketplace where all relevant information regarding medical equipment is available in one place. The platform aims to simplify the process of searching, comparing, and purchasing medical equipment while ensuring transparency in pricing and supplier authenticity. By integrating verified sellers, detailed product listings, and comparison tools, MedRex provides a streamlined solution that supports better decision making and improves overall efficiency in the healthcare procurement ecosystem.

Methodology

Our system is neither too large nor too small. We can deliver value in multiple stages (Product browsing, product comparison, and seller services). So we decided to use the Agile Method for our project. Some of the reasons for choosing this methodology are as follows:

- ✚ Customer satisfaction by rapid, continuous updates of medical equipment listings and platform features.
- ✚ People and interactions are emphasized rather than strict processes and tools. Customers, developers, and sellers continuously interact with each other to improve system quality.
- ✚ Close and continuous cooperation between users, verified sellers, and system maintainers to ensure accurate product information and pricing.
- ✚ Continuous attention to technical excellence and good system design to ensure reliability, scalability, and security in handling sensitive procurement data.
- ✚ Regular adaptation to changing requirements in the medical equipment market, including new product categories, pricing updates, and seller onboarding needs.
- ✚ To implement a new feature, developers need to lose only the work of a few days, or even only hours, to roll back and implement improvements without affecting the entire system.

System Description

System Request:

Project sponsors: Self sponsored.

Business Need:

To help healthcare institutions such as clinics, hospitals, and diagnostic laboratories search, compare, and choose among the best available medical equipment and procurement deals through a centralized digital marketplace.

Business Requirements:

Healthcare providers often face difficulties in collecting accurate and detailed information about medical equipment available in the market, including specifications, pricing, and supplier authenticity. MedRex is designed to solve this problem by providing a unified platform that enables users to efficiently search, compare, and evaluate medical equipment based on their specific requirements. The business requirements for the project “Medical Equipment Procurement and Marketplace System (MedRex)” are given below:

- ✚ A centralized database containing detailed information about all medical equipment products, organized using unique product identifiers.
- ✚ Search functionality for medical equipment based on specific user requirements such as category, price range, technical specifications, and availability. Users can input filters, and the system will display matching products from the database.
- ✚ Comparison feature allowing users to compare multiple medical equipment items based on various attributes such as specifications, functionality, price, and manufacturer details.
- ✚ Highlighting of best available deals based on pricing, ratings, and supplier reliability to assist users in making informed procurement decisions.
- ✚ User review and rating system for individual products to help other users evaluate product quality and performance based on real-world feedback.
- ✚ A user friendly interface that ensures efficient access to information, allowing healthcare professionals to obtain necessary procurement details without needing physical market visits or manual verification.

Business Value:

Since the exact revenue model and long term financial performance of MedRex cannot be precisely determined at this stage, a preliminary estimation has been developed based on expected platform usage, supplier engagement, and advertising opportunities.

- ✚ 400,000 BDT monthly revenue from verified medical equipment suppliers and distributors who will use the platform to list and promote their products.
- ✚ 120,000 BDT monthly revenue from targeted advertisements displayed on product pages and category sections within the platform.
- ✚ 250,000 BDT monthly operational cost including employee salaries, technical maintenance, and system support services.
- ✚ 180,000 BDT monthly infrastructure cost including office rent, cloud hosting, utilities, and security services required for stable platform operation.

Overall, the platform is expected to generate sustainable revenue growth as more healthcare institutions and verified suppliers join the system, increasing transaction volume and advertisement demand over time.

Special Issues or Constraints:

- ✚ The MedRex system is a moderately large scale platform and requires a minimum development period of approximately two months to ensure proper implementation of all core modules, including authentication, product management, order processing, and review systems. Each module must be carefully designed, tested, and integrated to ensure system reliability and performance.
- ✚ Since the platform is user centric and operates in a real-world healthcare procurement environment, continuous interaction with end users such as clinics, hospitals, and suppliers is necessary. Frequent feedback collection and iterative improvements are essential to refine usability, accuracy, and system efficiency.
- ✚ Data accuracy and trust are critical challenges, as incorrect or outdated product information may directly affect healthcare procurement decisions. Therefore, strict validation and seller verification processes must be maintained throughout system operation.
- ✚ The system must also handle scalability challenges as user traffic and product listings increase over time, requiring efficient database optimization and server side performance tuning.
- ✚ Ensuring cybersecurity is another major constraint, as the platform will store sensitive user and business data. Strong security mechanisms must be continuously maintained to prevent data breaches and unauthorized access.

Feasibility Analysis and Cost Calculation

The present value, Return of Investment and Break-Even point using sample data:

	Year 0	Year 1	Year2	Year 3	Total
Total Benefits	0	40,000	90,000	150,000	280,000
Total Cost	70,000	10,000	15,000	20,000	115,000
Net Benefits (Total Benefits – Total Cost)	(70,000)	30,000	75,000	130,000	165,000
Cumulative Net Cash Flow	(70,000)	(40,000)	35,000	165,000	

Financial Indicators:

ROI (Return on Investment):

$$= \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}} = \frac{280,000 - 115,000}{115,000} = 143.48\%$$

BEP (Break-Even Point):

The project breaks even during Year 2.

$$= \text{Year} + \frac{\text{Negative Cash Flow at start of Year}}{\text{Cash Flow During Year}} = 1 + \frac{40,000}{75,000} = 1.53 \text{ years}$$

Function Point Estimation

This table estimates the software size based on the features you described (Search, Comparison, Seller Accounts, Reviews).

Description	Total Num	Low (x)	Medium (x)	High (x)	Total
Inputs (Forms, Login, Add Product)	12	6 * 3	4 * 4	2 * 6	46
Outputs(Reports,Receipts,Comparison view)	8	3 * 4	3 * 5	2 * 7	41
Queries (Search, Filter, View Favorites)	15	8 * 3	5 * 4	2 * 6	56
Files (Database tables like Users, Products)	10	4 * 7	4 * 10	2 * 15	98
Programs/Interfaces (Payment API, License check)	4	2 * 5	1 * 7	1 * 10	27
Total Unadjusted Function Points (TUFPP)					268

Complexity and Final Score

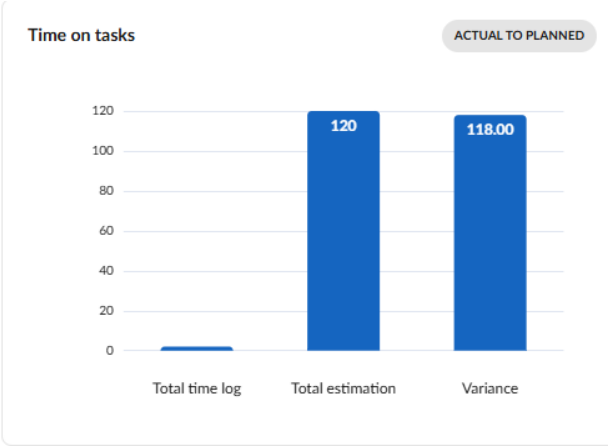
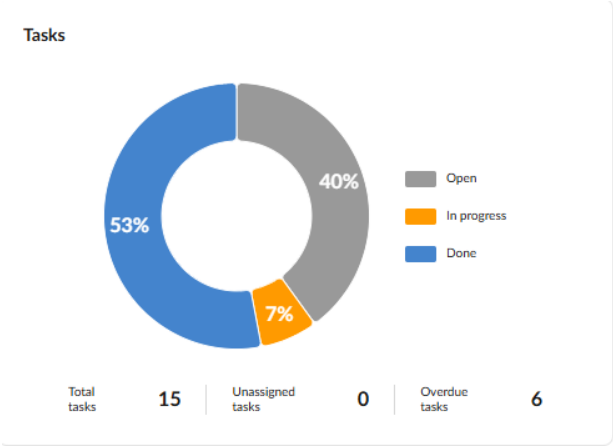
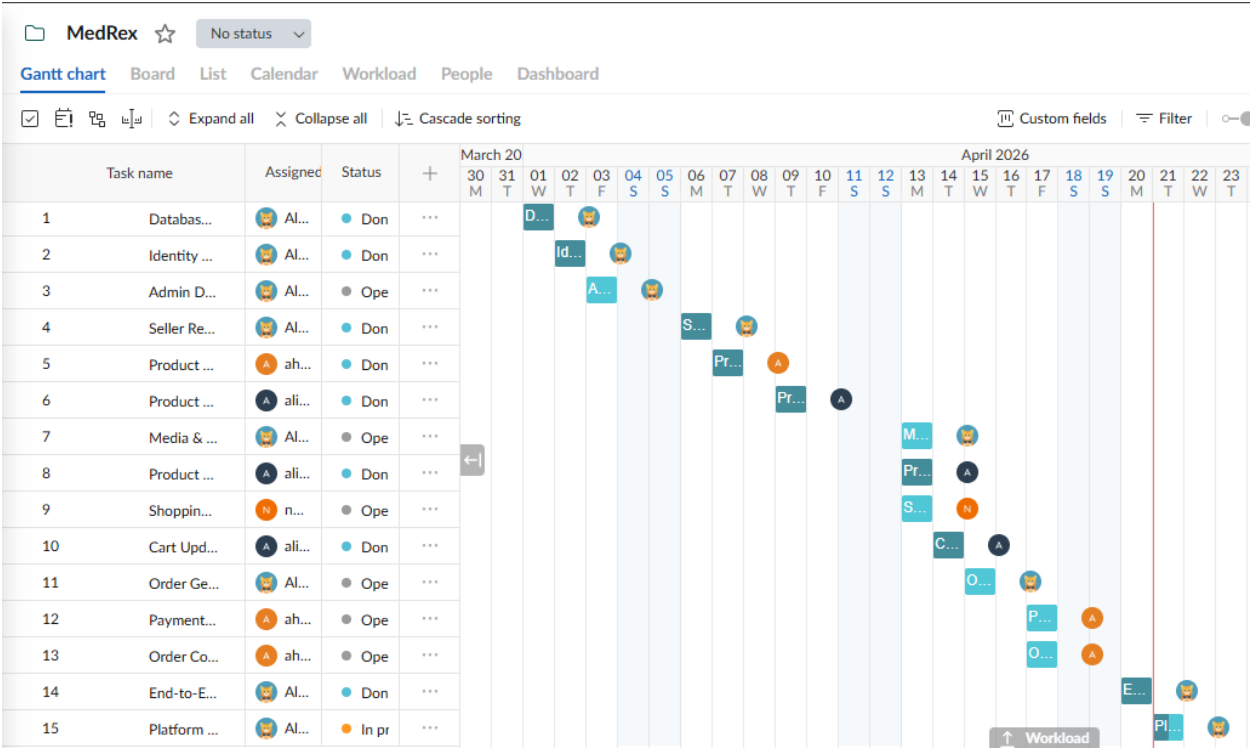
To get the Total Adjusted Function Points (TAFP), we look at your specific technical requirements (High security, fast response time).

- ✚ Data Entry: 4 (Extensive for Sellers)
- ✚ Performance: 5 (You required < 10ms response time)
- ✚ Online Update: 4 (Real-time availability updates)
- ✚ Total Processing Complexity: 13

Final Calculation:

- APC (Adjustment Process Complexity): $0.65 + (0.01 * 13) = 0.78$
- TAFP (Total Adjusted Function Points): $268 * 0.78 = 209.04$
- Estimated Size (Lines of Code) = $209.04 * 67 = 14,006$ Lines of Code
- Estimated Effort (Person-Months) = $209.04 / 17.5 = 11.95$ Person-Months
- Estimated Cost = $11.95 * 700 = 8,365$ BDT

Gantt Chart:



Functional Requirements:

- ✚ Database of all medical equipment with availability information: The system will maintain a comprehensive database of medical equipment sourced from global manufacturers. In addition to storing detailed specifications, the database must also include information regarding the availability of each product within the local market. This ensures that users can easily identify whether a particular piece of equipment can be procured within their region.
- ✚ Comparing devices: A comparison feature must be available where users can select multiple medical equipment items (up to a defined limit) and compare their specifications side by side. The comparison mode should include filtering options that allow users to focus on specific attributes such as functionality, technical specifications, price, category, and manufacturer details. Additionally, the system should include a highlighting mechanism to indicate the best-performing or most suitable option among the compared items, where applicable.
- ✚ Seller Accounts: The system must support a secondary type of user account known as “Seller account,” where verified medical equipment suppliers can register. These accounts must allow sellers to add new products, update existing listings, mark products as available for sale, and specify pricing information for each item.
- ✚ Specific Searches: Users must be able to search for medical equipment using specific criteria such as product name, category, price range, specifications, and brand. The system should return results that best match the user’s input requirements.
- ✚ Suggestions: The platform should display recommended products, including newly added equipment and best-selling items, to help users stay informed about current market trends and popular choices.
- ✚ Reviews and Rating: Users must be able to provide reviews and ratings for individual products. This feature will help future users gain insights into product quality, performance, and reliability based on real user experiences.

Non-Functional Requirements:

Product requirements

- a. Data encryption – The system must ensure secure storage of sensitive data using encryption and hashing techniques to prevent unauthorized access.
- b. < 10ms response time – The system should provide fast performance with response time under 10 milliseconds for core operations like search and browsing.
- c. Seller account verification – Only verified sellers with valid credentials are allowed to access seller features and publish products.

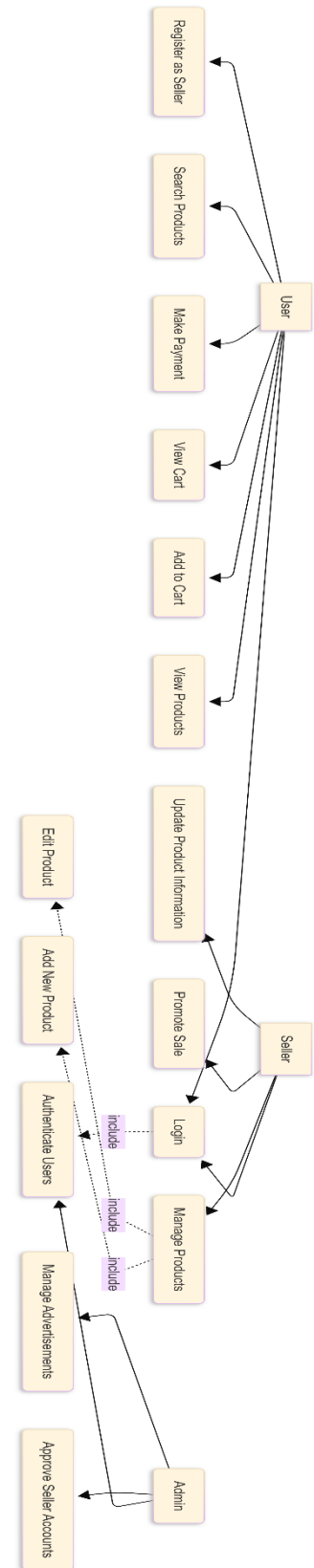
Organizational requirements

- a. PHP on Server – The backend system is implemented using PHP for server-side processing.
- b. MySQL for database – MySQL is used as the primary database to store and manage all system data.
- c. Seller license requirement – Seller accounts must be registered with valid business licenses to ensure authenticity and trust.

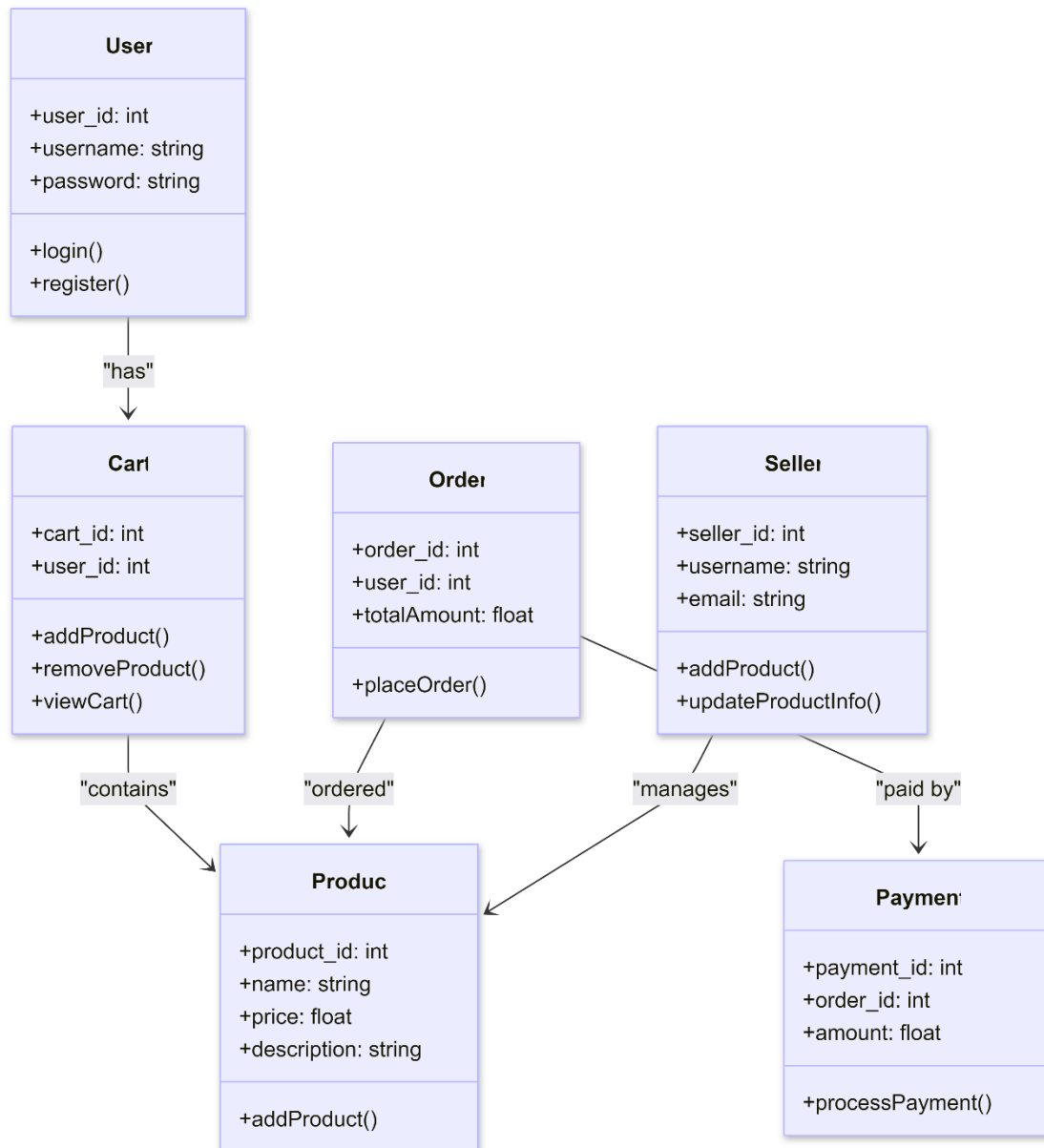
System Design

Use Case Diagram:

The system functions as a multi-tier e-commerce platform where Users can search for and view products, manage their cart, and make payments, with the option to register as a Seller. Sellers maintain control over the marketplace by managing, adding, and editing product listings, updating information, and promoting sales, while the Administrator oversees the platform by managing advertisements and approving new seller accounts to ensure site integrity. Crucially, all sensitive actions including logging in and product management include a mandatory Authenticate Users step, ensuring that every transaction and inventory change is performed by a verified actor.

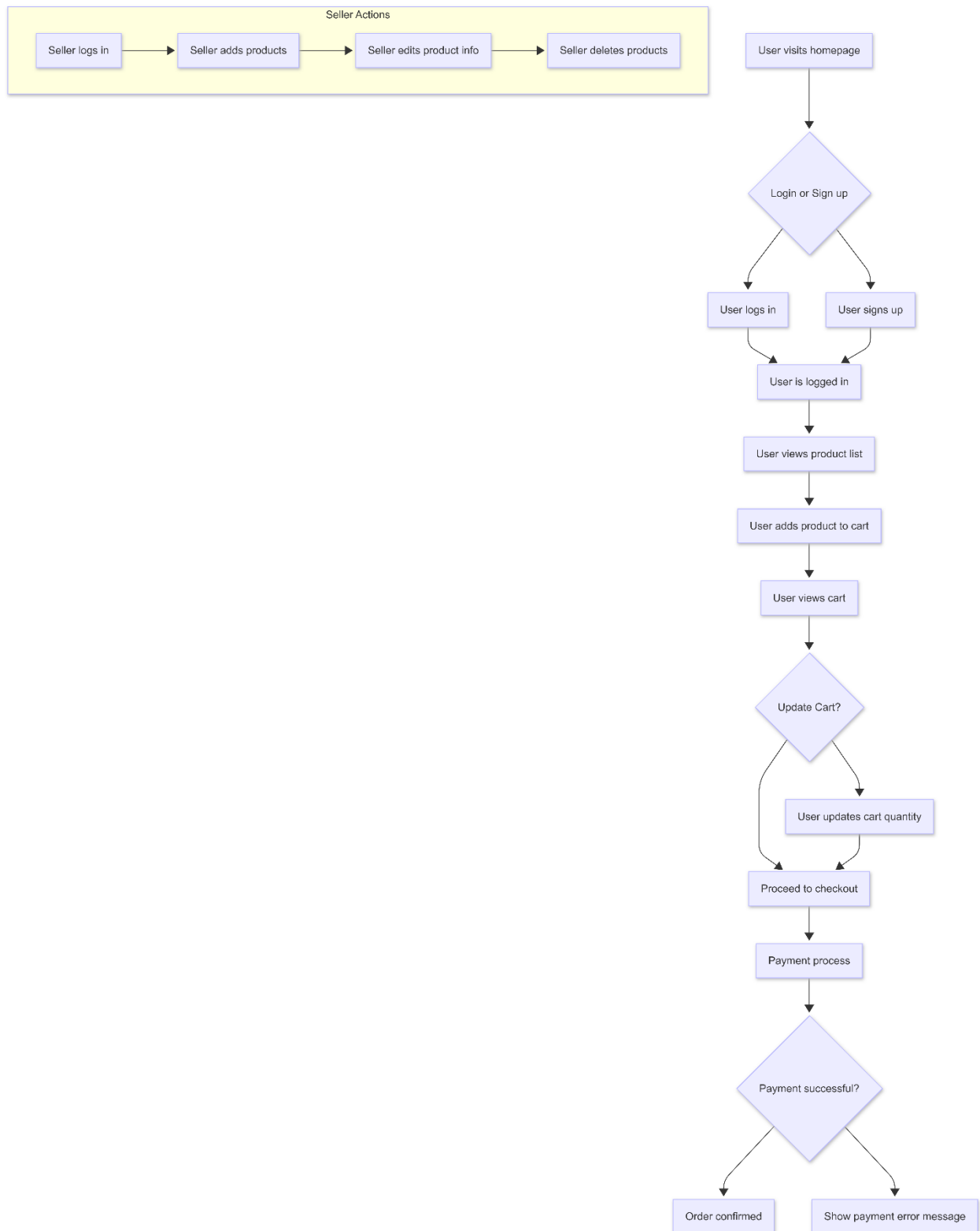


Class Diagram



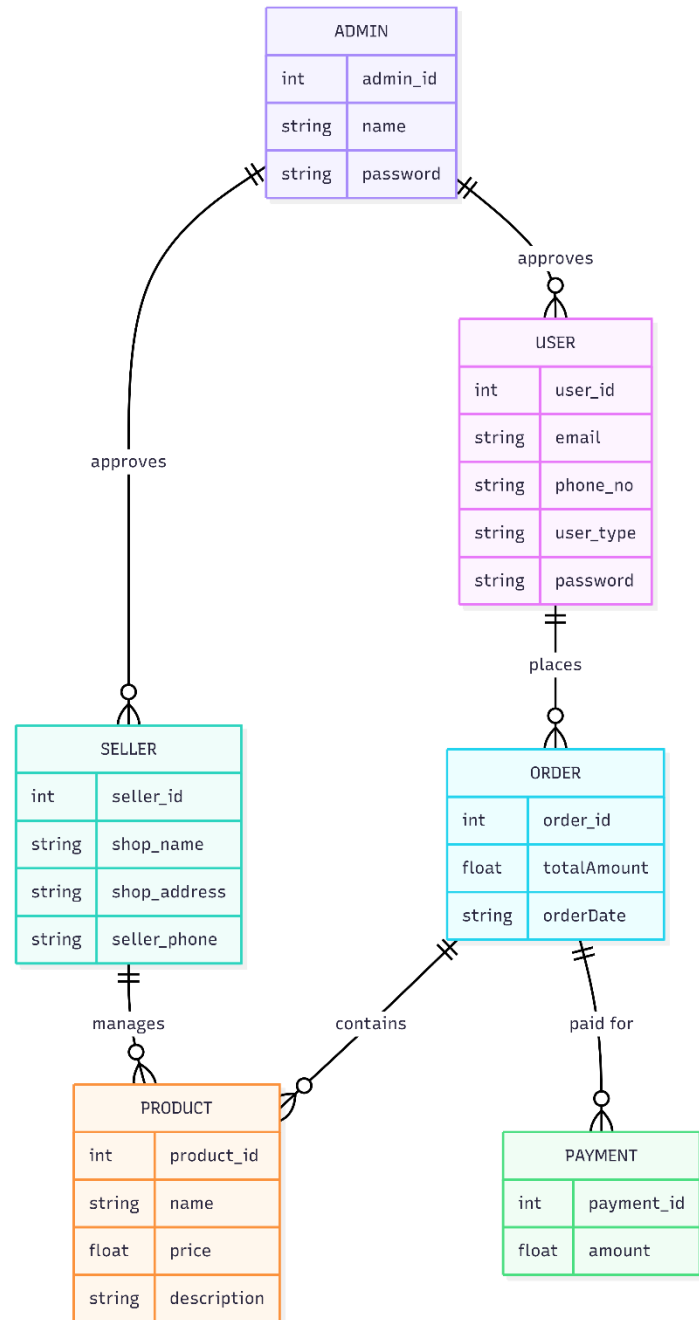
The system architecture is structured around the User and Seller actors, where a User has a Cart to manage product selections and an Order class to finalize purchases through the placeOrder() method. The Seller manages the Product inventory by adding or updating item details and is directly associated with the Payment class to process transactions for completed orders. All core entities including Cart, Order, and Payment interact with the Product class, ensuring that data such as price and product_id flows consistently from the initial search to the final payment processing stage.

Activity Diagram



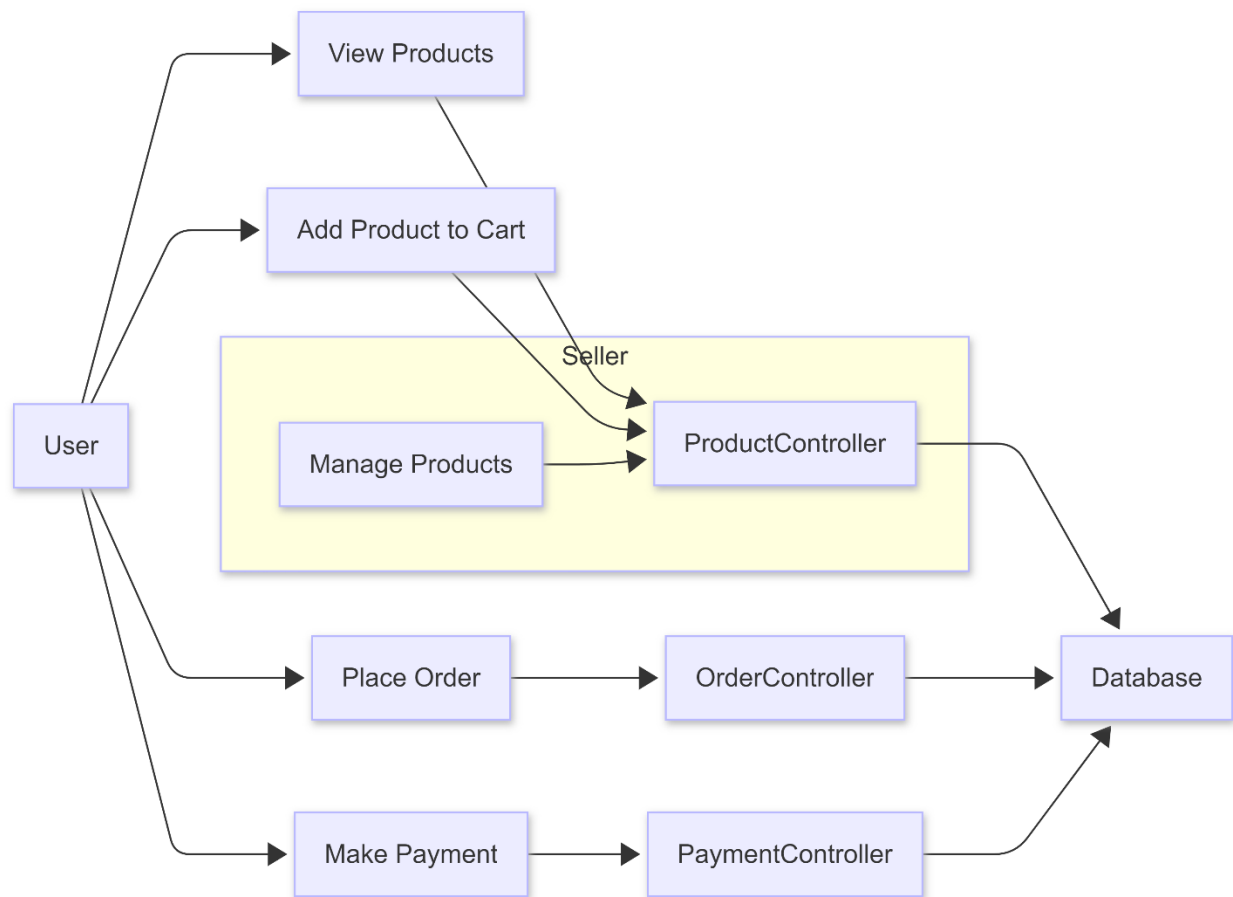
In the activity diagram, we assume that the operational workflow begins with a decision point where a user must either log in or sign up to access core features, while the Seller Actions are sequestered into a specialized flow for adding, editing, and deleting product information. Once authenticated, the user follows a linear progression from viewing the product list to adding items to their cart, with a specific decision node allowing them to update cart quantities before proceeding to checkout. The final phase of the activity focuses on the payment process, which branches into two terminal states based on a success condition: either culminating in an "Order confirmed" status or triggering a "Show payment error message" to notify the user of a failed transaction.

Entity Relationship Diagram (ERD)



The Entity Relationship Diagram (ERD) defines the data structure of the platform, where a central ADMIN manages the integrity of the system by approving multiple USER and SELLER accounts through a one-to-many relationship. A USER can place multiple ORDERS, each of which is "paid for" by a specific PAYMENT entry and "contains" various items from the PRODUCT entity. On the supply side, a SELLER manages multiple PRODUCTS, ensuring that each item listed in the database is tied to a specific shop's details, such as shop_name and shop_address, while the overall schema maintains referential integrity through unique identifiers like user_id, seller_id, and order_id.

Data Flow Diagram:



In the Data Flow Diagram (DFD), we assume that the information flow is managed through specialized controllers where the User interacts with the system to view products, add items to their cart, place orders, and make payments, with each request being processed by the ProductController, OrderController, or PaymentController respectively. The Seller maintains the core inventory by managing products through the same ProductController, ensuring that all updates are systematically validated before reaching the central Database. This architecture ensures that data flows seamlessly from the external entities through the processing logic and is securely stored or retrieved from the persistent storage layer, maintaining consistent record-keeping for every transaction and inventory change.

Timeline:

29/03/2026 – Started discussion of the system

29/03/2026- Completed data gathering and requirements

17/04/2026- The system has been completed

20/04/2026- Done with all the necessary testing

21/04/2026- System is ready to launch

References:

[Tools]

[1] GanttPro for Gantt Chart

[2] Marmaid for Use case,ERD,Class,DFD and Activity Diagram.